Hypothesis testing on antimicrobial zone-of-inhibition data

A Python project that compares plant extracts with a reference antibiotic using parametric and non-parametric tests, and uses simulation to show Type I error, multiple testing and statistical power.

The data in this project is simulated. I generated zone-of-inhibition measurements for four groups so I could practise the full testing workflow on a problem where I know how the data were made. The workflow is the same for real plate measurements.

Scenario

A lab tests three plant extracts (A, B, C) and a reference antibiotic against a bacterial culture. Each is tested on 12 agar plates, and the clear ring around each disc (the zone of inhibition, in mm) is measured. A bigger zone means stronger antimicrobial activity.

Questions:

1. Do the four groups differ in mean zone size?
2. Which groups differ from each other?
3. How reliable are these conclusions, and how many plates would a better test need?

Method

1. **Check assumptions.** Shapiro-Wilk for normality in each group and Levene's test for equal variances.
2. **Compare all four groups.** One-way ANOVA (parametric) and Kruskal-Wallis (non-parametric), so I can see whether the choice of test changes the answer.
3. **Follow up on pairs.** Tukey HSD after ANOVA, and Mann-Whitney U with Holm correction. Both correct for running six comparisons.
4. **Look at one pair in detail.** Extract B vs the reference antibiotic, with Welch's t-test, a 95% confidence interval and Cohen's d.
5. **Simulate error rates.** Repeat experiments 5,000 times to measure the Type I error rate, the false alarm rate across six tests, and power at different sample sizes.

Results

Group	Mean zone (mm)	SD
Extract A	12.9	2.0
Extract B	13.9	2.0
Extract C	12.5	1.4
Reference antibiotic	18.3	1.5

- **Assumptions:** Extract C is not normally distributed (Shapiro-Wilk $p = 0.002$), and the other groups look normal. Variances are similar (Levene $p = 0.39$).
- **Overall difference:** ANOVA gives $F = 27.6$, $p < 0.001$, with eta squared = 0.65, meaning group explains about 65% of the variation. Kruskal-Wallis agrees ($H = 27.5$, $p < 0.001$). Since both tests give the same answer, the non-normal Extract C does not change the conclusion.
- **Which groups differ:** the reference antibiotic is significantly higher than all three extracts. The extracts do not differ significantly from each other (all adjusted p-values above 0.2).
- **Extract B vs reference:** a difference of 4.4 mm (95% CI 2.9 to 5.9 mm), Welch $t = -6.18$, $p < 0.001$, Cohen's $d = 2.5$, a very large effect.

Error rates and power (simulation)

- A single test at alpha = 0.05 gives a false alarm 4.9% of the time, as it should.
- Running all six pairwise comparisons without correction gives at least one false alarm 19.9% of the time. With Holm correction this falls to 4.0%.
- To detect a real 2 mm difference (SD 2 mm), 10 plates per group gives only 56% power, and 12 plates gives 65%. About 17 plates per group are needed for 80% power.

What this means

The extracts are clearly weaker than the reference antibiotic. But the extracts were not significantly different from each other, and with 12 plates per group the experiment was underpowered to detect a 2 mm difference. "No significant difference" here is not evidence that the extracts are equal. A follow-up experiment with about 17 plates per group would answer that question properly.

Limits

- The data are simulated, so the results show the method, not a biological finding.
- Twelve plates per group is small, and the Shapiro-Wilk test has low power at this size, so a normal-looking group could still be non-normal.
- I assumed independent plates. Real plates from the same batch or day would need a model that accounts for that.

Files

File	What it is
`hypothesis_testing_analysis.py`	The full analysis, from simulation to figure
`zone_of_inhibition.csv`	The simulated measurements
`test_results.csv`	Every test, with statistic and p-value
`hypothesis_testing.png`	Boxplot, power curve and false alarm chart

Run it

```
pip install numpy pandas scipy matplotlib
python hypothesis_testing_analysis.py
```

The random seed is fixed, so you get the same numbers every time.